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DecAug: Out-of-Distribution Generalization via Decomposed Feature Representation and Semantic Augmentation

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Out-of-distribution generalization

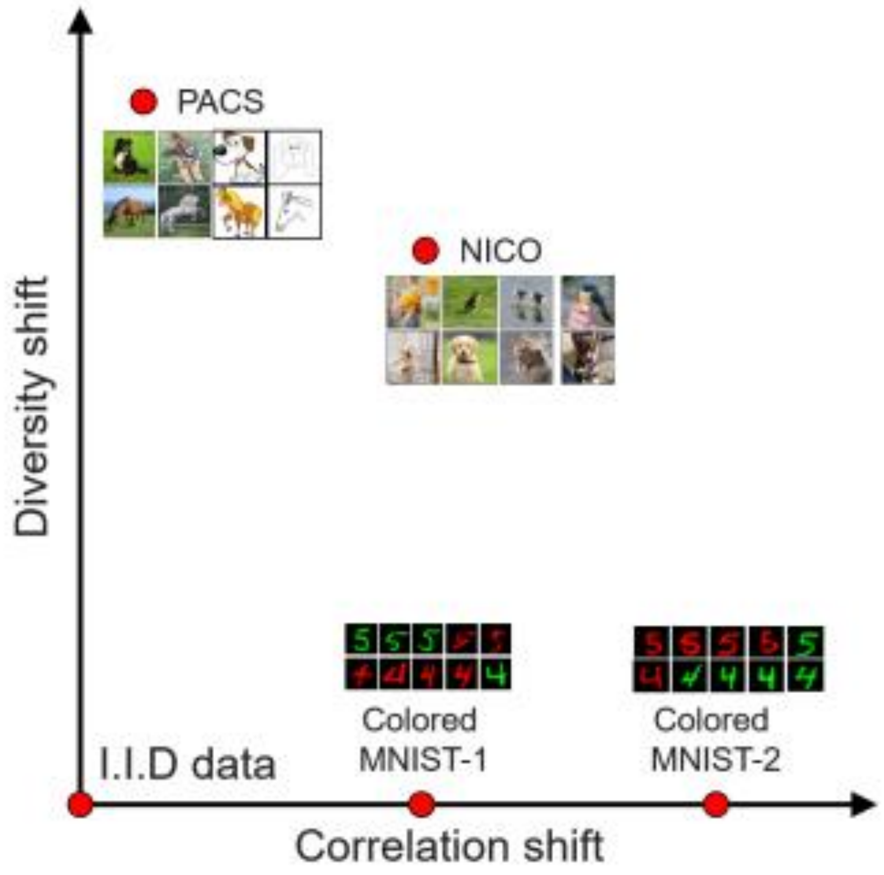

Figure 1: Illustration of the two-dimensional out-of-distribution problem.

- **Objective:**
  Out-of-distribution (OoD) generalization

- **Two dimensional OoD:**
  Diversity shift and correlation shift

Two dimensional OoD

Figure 2: Typical examples of out-of-distribution correlation shift data.

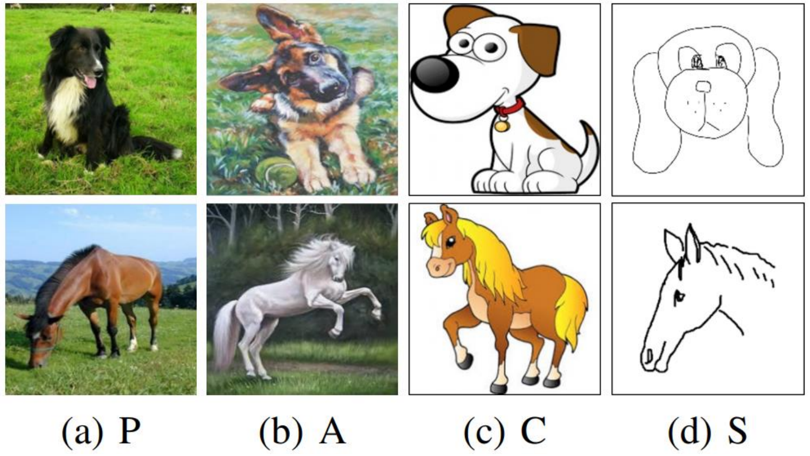

Figure 3: Typical examples of out-of-distribution diversity shift data.

- ➢ We identify two types of OoD factors: correlation shift and diversity shift.
- ➢ Extensive experiments show that many OoD methods can only solve one dimension of OoD generalization.

Figure 4: Some examples of the two-dimensional out-of-distribution data.

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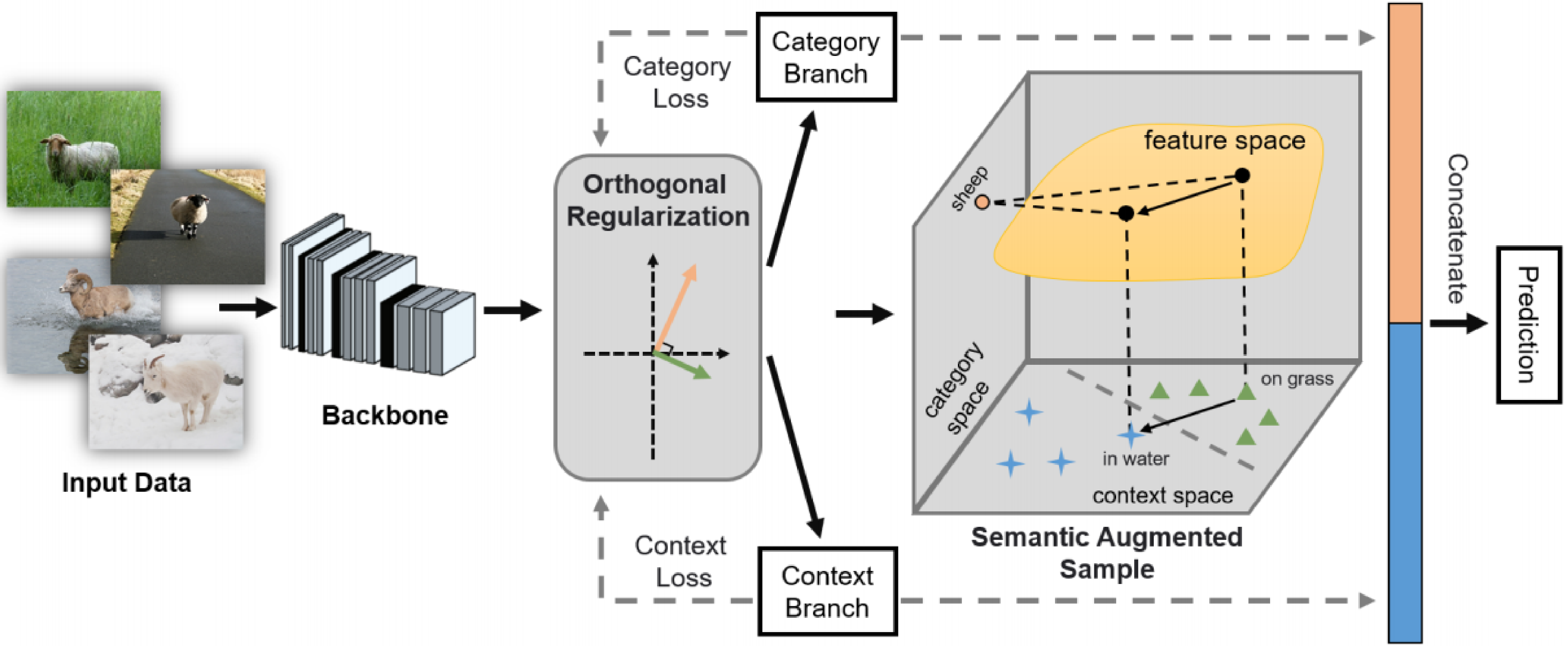

Figure 5: An overview of the proposed DecAug.

➢ **Method**

DecAug, A novel **dec**omposed feature representation and semantic **aug**mentation approach for OoD generalization.

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Algorithm 1 DecAug: Decomposed Feature Representation and Semantic Augmentation for OoD generalization

Input: Training set $\mathcal{D}$, batch size n, learning rate β, hyper-parameters ϵ, λ^1, λ^2, λ^{orth}.
Output: $\theta, \theta^1, \phi^1, \theta^2, \phi^2, \phi$.

Initialize $\theta, \theta^1, \phi^1, \theta^2, \phi^2, \phi$;
repeat
  Sample a mini-batch of training images $\{(x_i, y_i, c_i)\}_{i=1}^{n}$ with batch size n;
  for each (x_i, y_i, c_i) **do**
    $z_i \leftarrow g_\theta(x_i)$;
    $\mathcal{L}_i^1(\theta, \theta^1, \phi^1) \leftarrow \ell(h_{\phi^1} \circ f_{\theta^1}(z_i), y_i)$;
    $\mathcal{L}_i^2(\theta, \theta^2, \phi^2) \leftarrow \ell(h_{\phi^2} \circ f_{\theta^2}(z_i), c_i)$;
    Compute $\mathcal{L}_i^{\text{orth}}(\theta^1, \phi^1, \theta^2, \phi^2)$ according to Eq. (1);
    Randomly sample α_i from $[0, 1]$;
    Generate $\tilde{z}_i^2$ according to Eq. (2);
    $\mathcal{L}_i^{\text{concat}}(\theta, \theta^1, \theta^2, \phi) \leftarrow \ell(h_\phi([f_{\theta^1}(z_i), \tilde{z}_i^2]), y_i)$;
    Compute $\mathcal{L}_i(\theta, \theta^1, \phi^1, \theta^2, \phi^2, \phi)$ according to Eq. (3);
  end for
  $(\theta, \theta^1, \phi^1, \theta^2, \phi^2, \phi) \leftarrow (\theta, \theta^1, \phi^1, \theta^2, \phi^2, \phi) - \beta \cdot \nabla \frac{1}{n} \sum_{i=1}^{n} \mathcal{L}_i(\theta, \theta^1, \phi^1, \theta^2, \phi^2, \phi)$;
until convergence;

Pseudo code for DecAug.

- DecAug disentangles the category-related and context-related features.
- The decomposition is achieved by orthogonalizing the two gradients of losses for predicting category and context labels
- Gradient-based augmentation on context-related features to improve the robustness.

3 Results and discussion

Table 1: Results on the PACS benchmark.

Model	Art Painting	Cartoon	Sketch	Photo	Average
ERM (Carlucci et al. 2019)	77.85	74.86	67.74	95.73	79.05
IRM (Arjovsky et al. 2019)*	70.31	73.12	75.51	84.73	75.92
Rex (Krueger et al. 2020)*	76.22	73.76	66.00	95.21	77.80
JiGen (Carlucci et al. 2019)	79.42	75.25	71.35	**96.03**	80.51
DANN (Ganin et al. 2016)	81.30	73.80	74.30	94.00	80.80
MLDG (Li et al. 2017b)	79.50	77.30	71.50	94.30	80.70
CrossGrad (Shankar et al. 2018)	78.70	73.30	65.10	94.00	80.70
MASF (Dou et al. 2019)	80.29	77.17	71.69	94.99	81.03
Cumix (Mancini et al. 2020)	**82.30**	76.50	72.60	95.10	81.60
DecAug	*79.00*	***79.61***	***75.64***	*95.33*	***82.39***

* Implemented by ourselves.

Table 2: Results on the Colored MNIST.

Model	Acc test env
ERM*	17.10 ± 0.6
IRM (Arjovsky et al. 2019)	66.90 ± 2.5
Rex (Krueger et al. 2020)	68.70 ± 0.9
F-IRMGames (Ahuja et al. 2020)	59.91 ± 2.7
V-IRMGames (Ahuja et al. 2020)	49.06 ± 3.4
ReBias (Bahng et al. 2020)*	29.40 ± 0.3
JiGen (Carlucci et al. 2019)*	11.91 ± 0.4
DecAug	***69.60*** ± ***2.0***
ERM, grayscale model(oracle)	73.00 ± 0.4
Optimal invariant model (hypothetical)	75.00

Table 3: Results on the NICO dataset.

Model	Animal	Vehicle
ERM*	75.87	74.52
IRM (Ahuja et al. 2020)*	59.17	62.00
Rex (Krueger et al. 2020)*	74.31	66.20
Cumix (Mancini et al. 2020)*	76.78	74.74
DANN (Ganin et al. 2016)*	75.59	72.23
JiGen (Carlucci et al. 2019)*	84.95	79.45
CNBB (He, Shen, and Cui 2020)*	78.16	77.39
DecAug	***85.23***	***80.12***

➢ DecAug outperforms other state-of-the-art methods on various OoD datasets: Color-MNIST, PACS and NICO.

3 Ablation study

Table 4: Ablation study on PACS with ResNet-18.

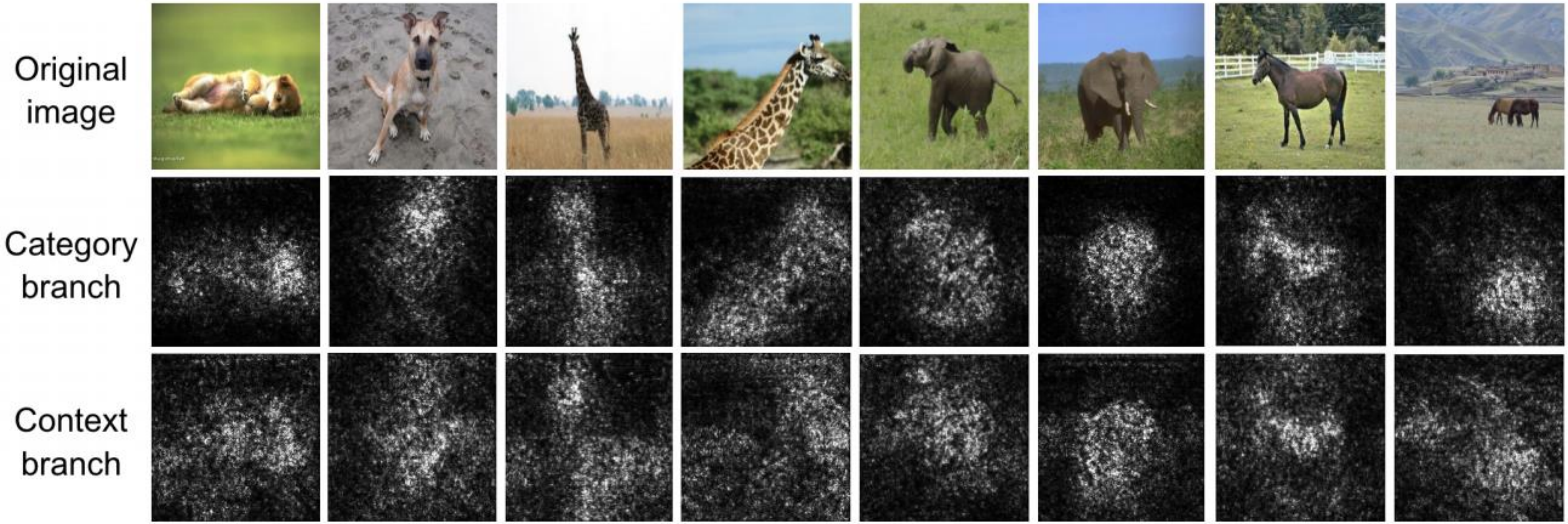
PACS	Art Painting	Cartoon	Sketch	Photo	Average
DecAug without orth loss	78.42	78.32	72.13	94.19	80.77
DecAug (orth 0.0005)	77.49	77.43	74.32	94.07	80.76
DecAug (orth 0.001)	78.12	77.34	76.97	94.91	81.83
DecAug (orth 0.01)	***79.00***	***79.61***	***75.64***	***95.33***	***82.39***

Table 5: Results of DecAug variants on the PACS dataset.

Model	Average
DecAug (DANN loss)	81.00
DecAug (orth between features)	79.90
DecAug (gradient-based orth)	***82.39***

- ➢ The ablation studies confirm the effectiveness of the proposed orthogonal loss.
- ➢ The two branches architecture is needed, and adversarial augmentation is better performed on the context predicting branch to improve OoD generalization.

3 Interpretability analysis



- The saliency maps of the category branch focus more on foreground objects, while the saliency maps of the context branch are also sensitive to background contexts that contain domain information.

4 Conclusion

➢ A novel decomposed feature representation and semantic augmentation method for various OoD generalization tasks:

- High level representations are decomposed into category-related and context-related features.

- Gradient-based semantic augmentation is performed on the context-related features.

- First method can achieve SOTA performance on various OoD generalization tasks from different research areas.

THANKS